

Introduction to Meshtastic

Mesh Networking for Off-Grid Communication



Agenda

Today's Topics

Why Use Meshtastic?

How does it work?

What Is Meshtastic?

Possible Scenarios

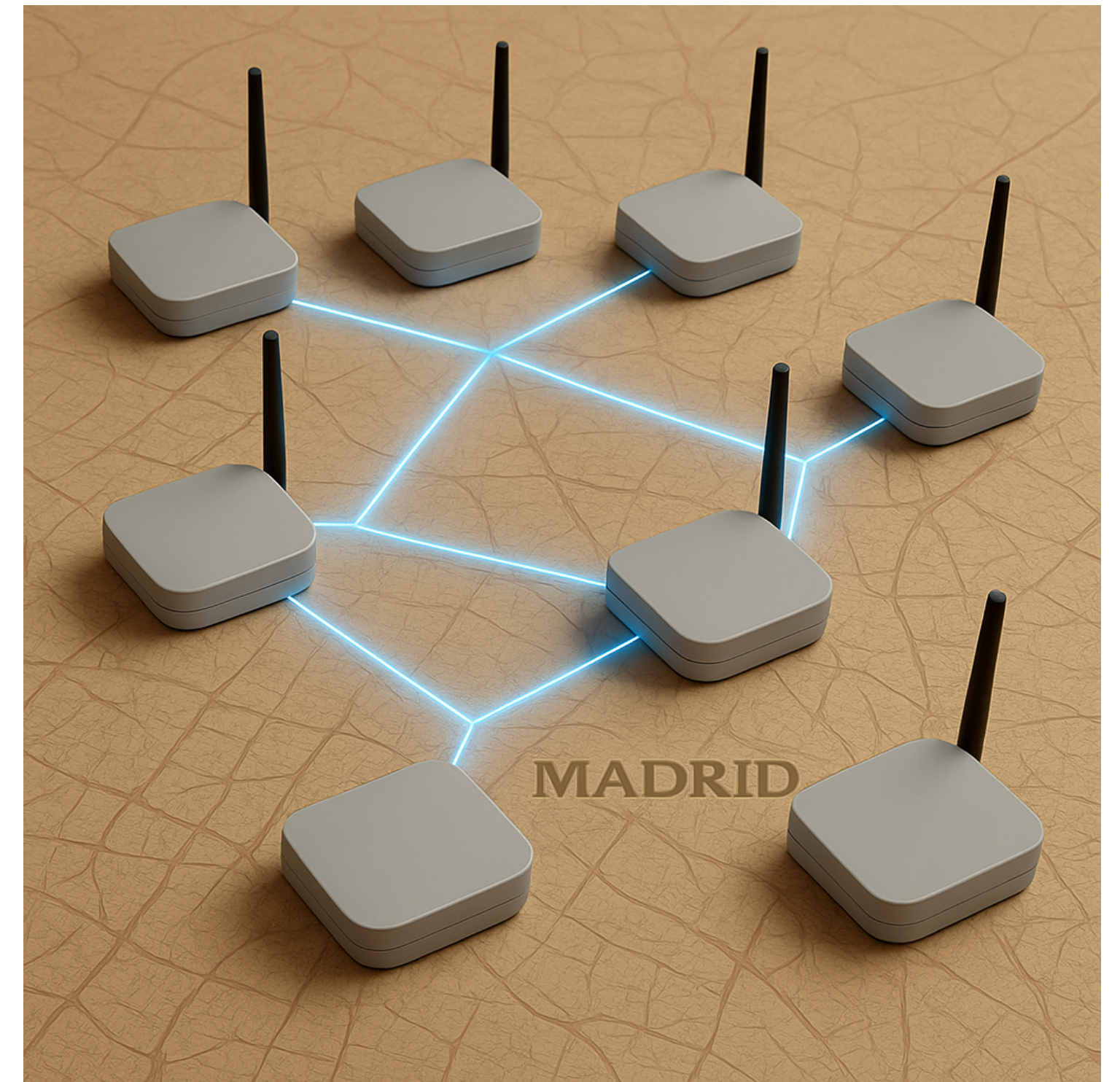
How to Start Using Meshtastic?



Why Use Meshtastic?

Imagine we could have:

- Off-grid text radio communication
- Long-range connectivity
- Low power consumption (solar)
- Very affordable hardware
- Free, no subscriptions or fees
- Broadcast network, different roles
- End-to-end encryption
- Totally legal



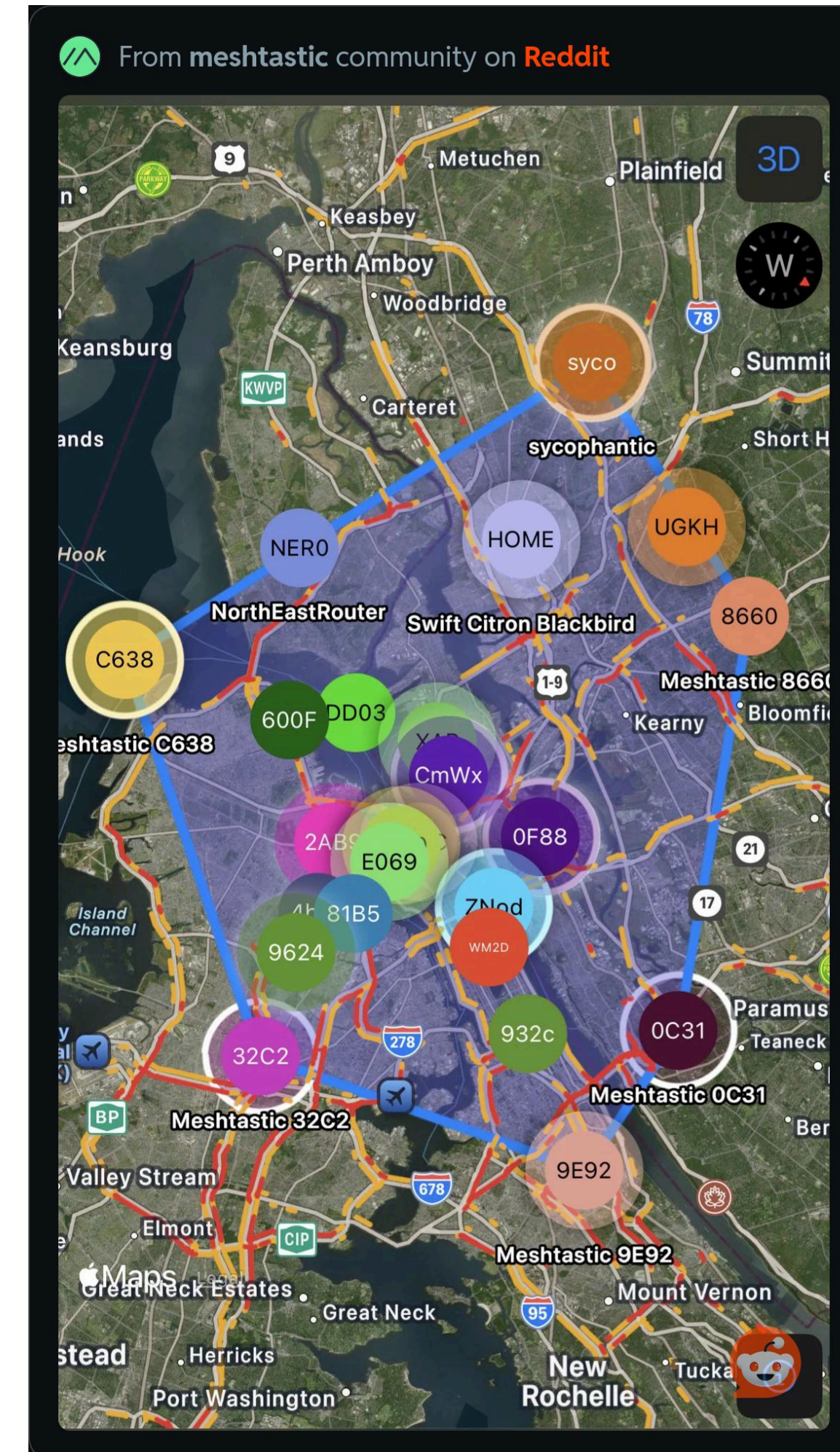
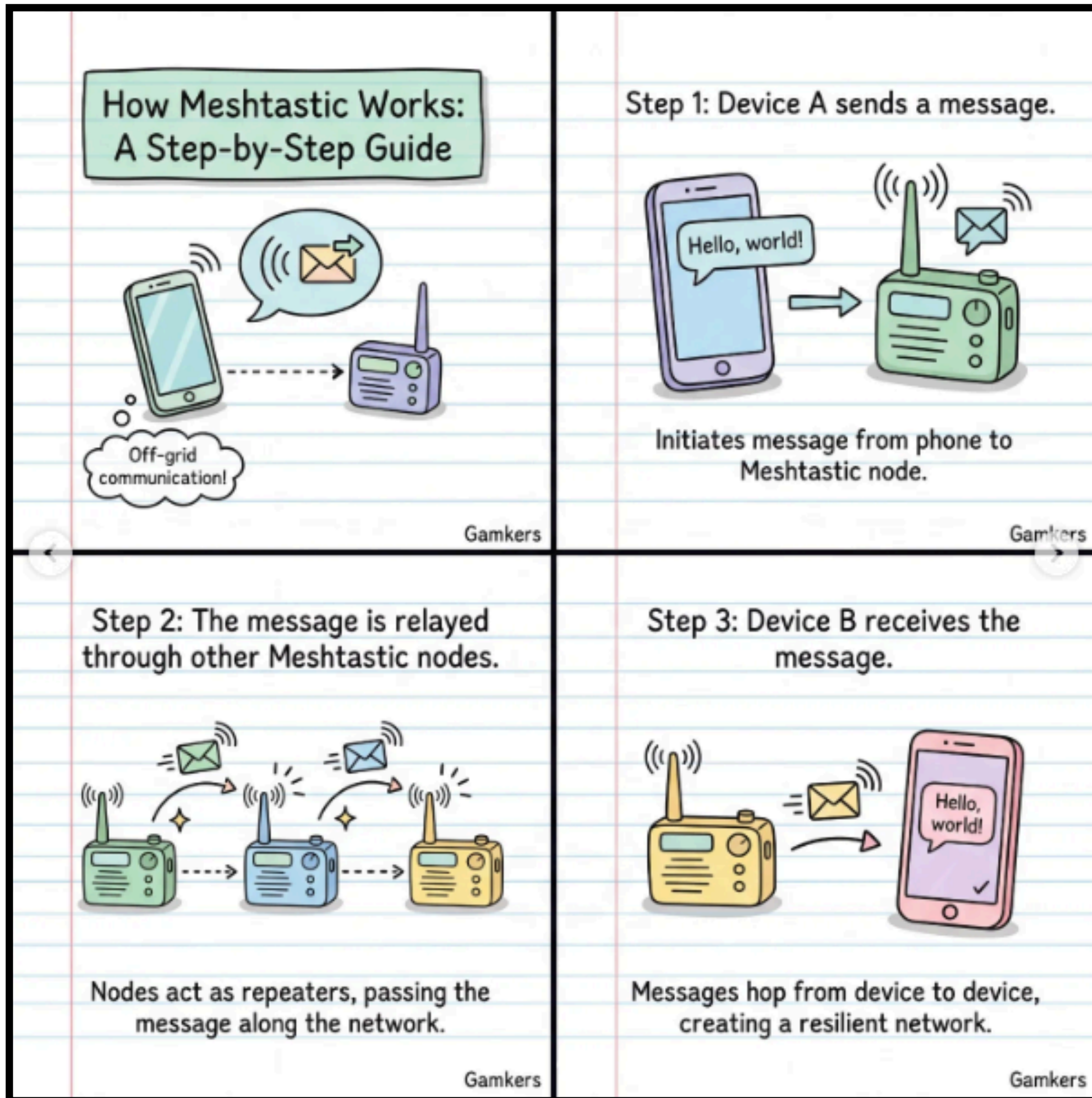
How does it work?

Class analogy



- Passing notes
- Creating copies of the note for other people
- In just a few hops full class is covered
- Encrypted just in case teacher finds out
- Messages can be public, or private

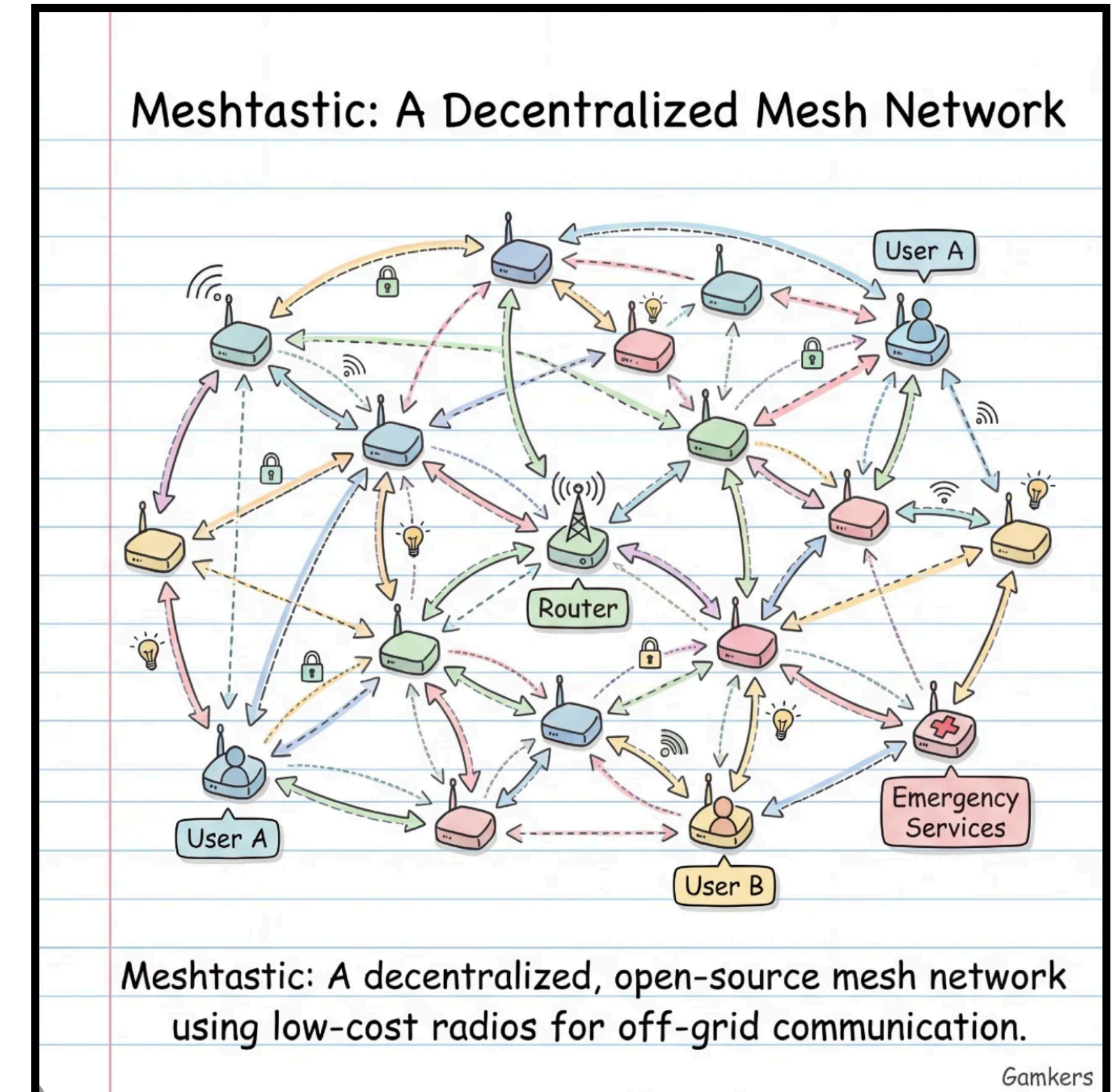
Passing messages around



What is Meshtastic?

Basics

- Open-source mesh networking project
- Many countries (80 plus)
- Uses LoRa radio technology
- Stand alone nodes or via smartphone app
- Peer-to-peer, no servers
- Flavors: Private, Groups, Public



Possible Scenarios

Use Cases

- Community mesh networks
- Outdoor & hiking, festivals, cruise
- Blackout or mobile operator issues
- Emergency response and disaster zones
- Remote area communication
- Telemetry and integrations with sensors, other networks and services (ATAK)
- Private communication



Possible Scenarios

Use Cases



Ayuntamiento de Ribadesella

PRINCIPADO DE ASTURIAS



Manual Comunitario de Comunicaciones de Emergencia

CB – PMR – VHF – Meshtastic

1. Objetivo

Este manual establece un protocolo de comunicación comunitaria en caso de que fallen las redes de telefonía e internet. Su fin es asegurar que la población reciba información clara y oficial mediante radio CB, PMR y VHF.

2. Canales designados

Banda	Canal / Frecuencia	Uso principal	Observaciones
CB-27 MHz	Canal 9 (AM)	Emergencias	Escucha continua
	Canal 3 (AM)	Boletines oficiales	Transmisión de comunicados
PMR446 (UHF)	Canal 1	Emergencias	Escucha continua
	Canal 3	Boletines oficiales	Transmisión de comunicados
VHF (radioaficionados)	145.500 MHz	Apoyo técnico y retransmisión	Operado por radioaficionados autorizados
Meshtastic	Ribadesella(*)	Boletines oficiales	Transmisión de comunicados

(*) Contraseña canal: AQ==

How to Start Using Meshtastic

Getting Started

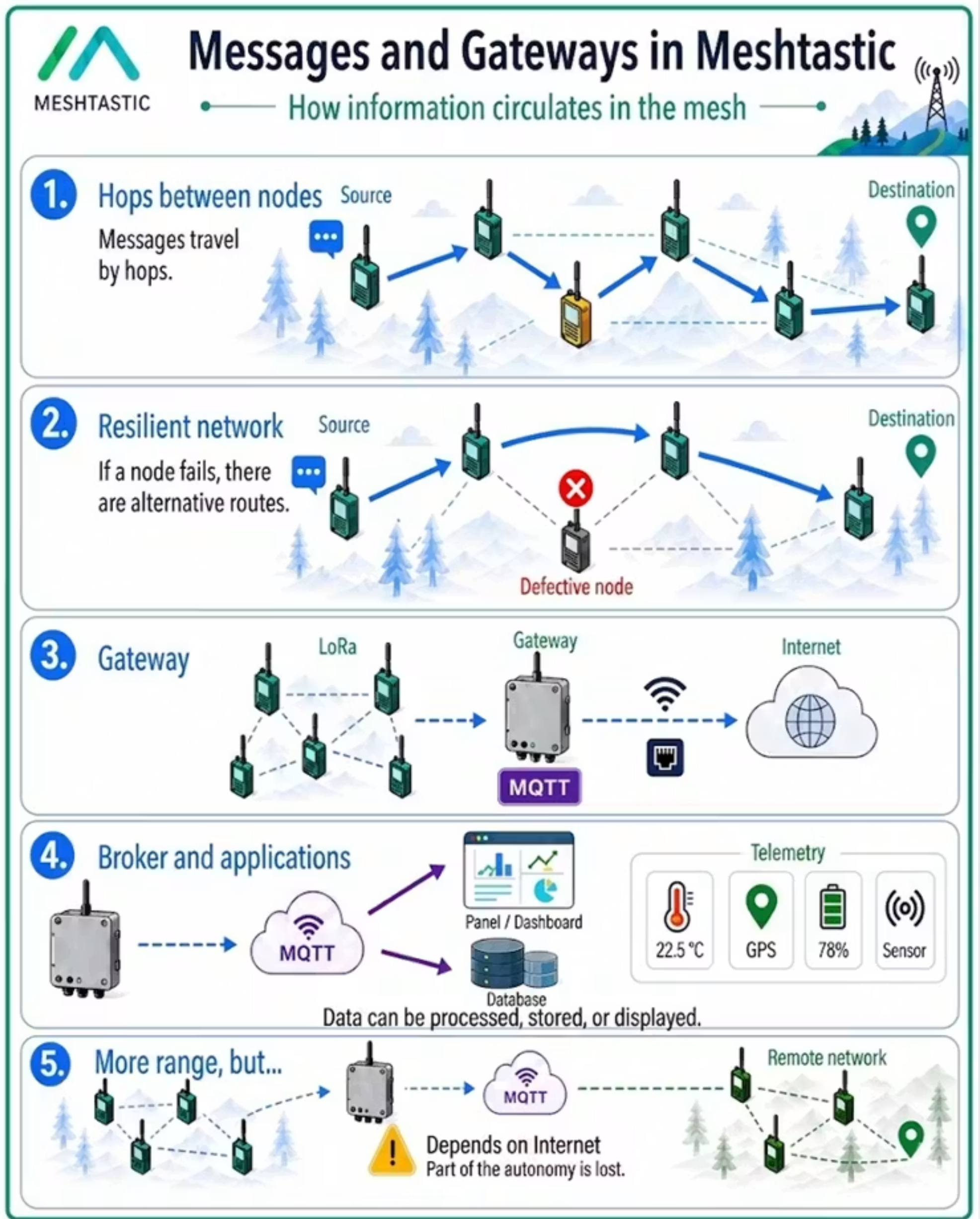
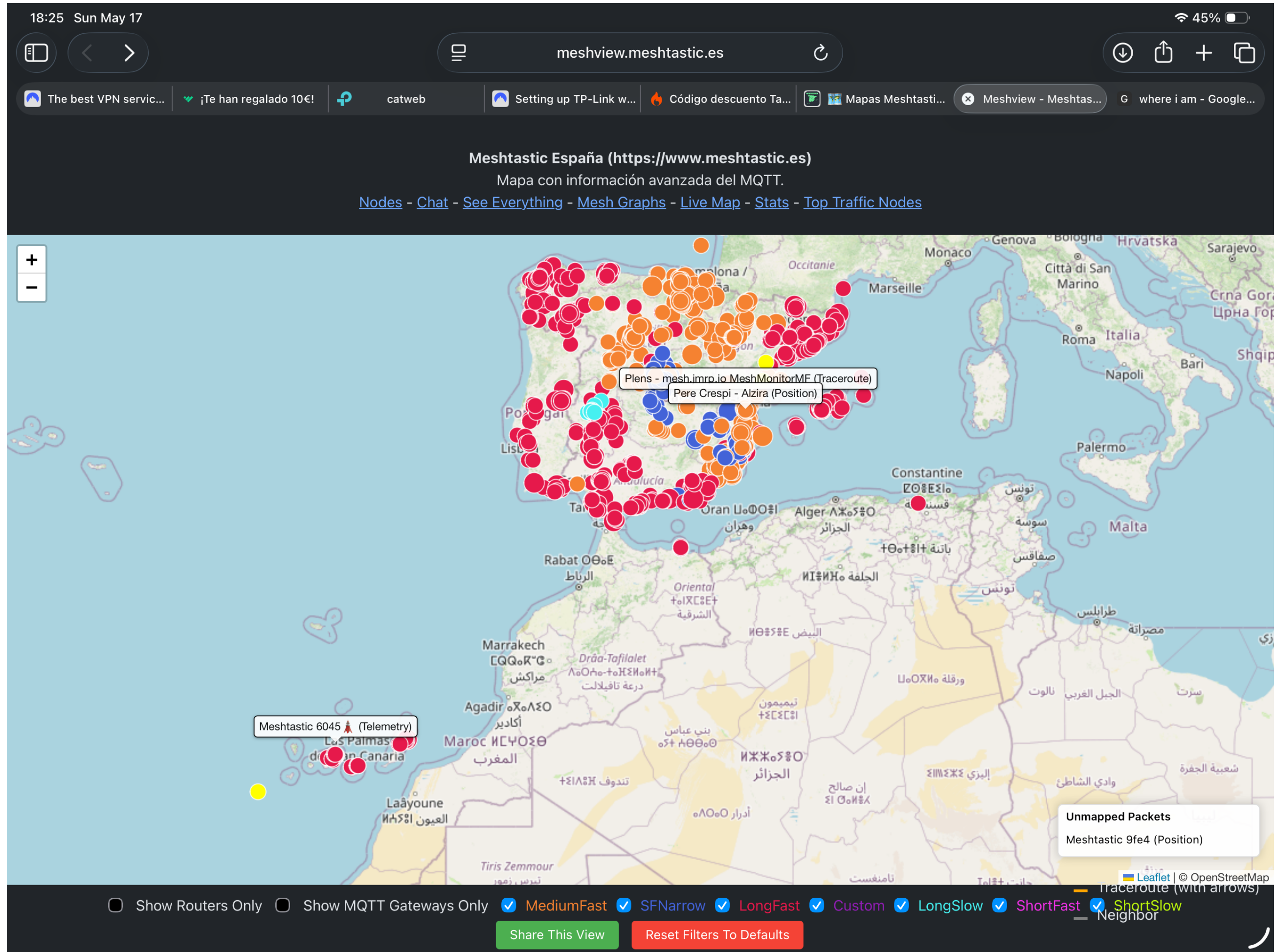


How to Start Using Meshtastic

Getting Started

- Purchase a “device”
 - Install Meshtastic firmware
 - Download mobile app
 - Configure node
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- Bonus: join Telegram or Reddit





From gadget to community

Change of mindset

**You are part of the
network for the
community to
communicate**



Questions?



Introduction to Meshtastic

Thanks!

